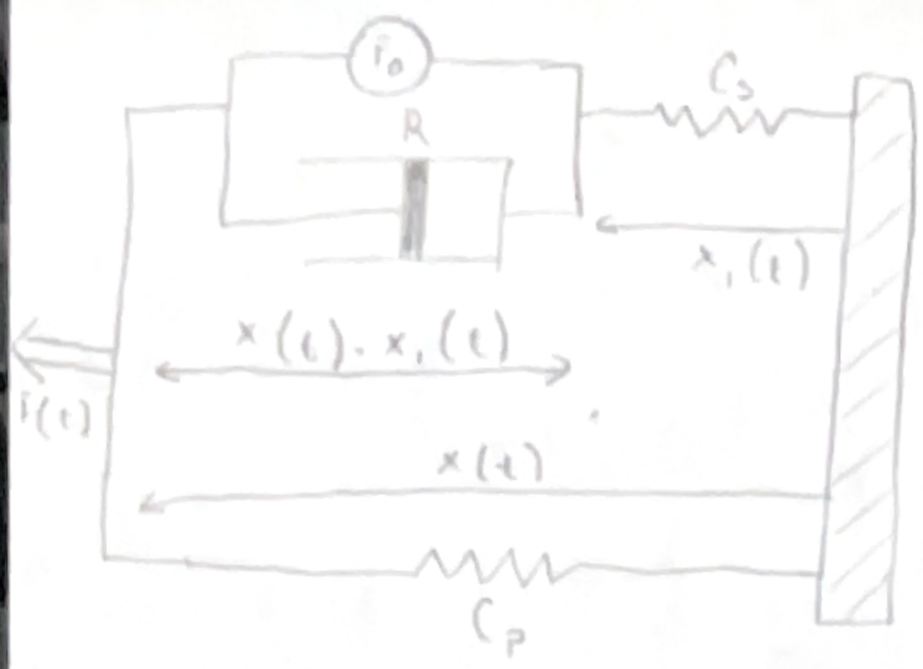
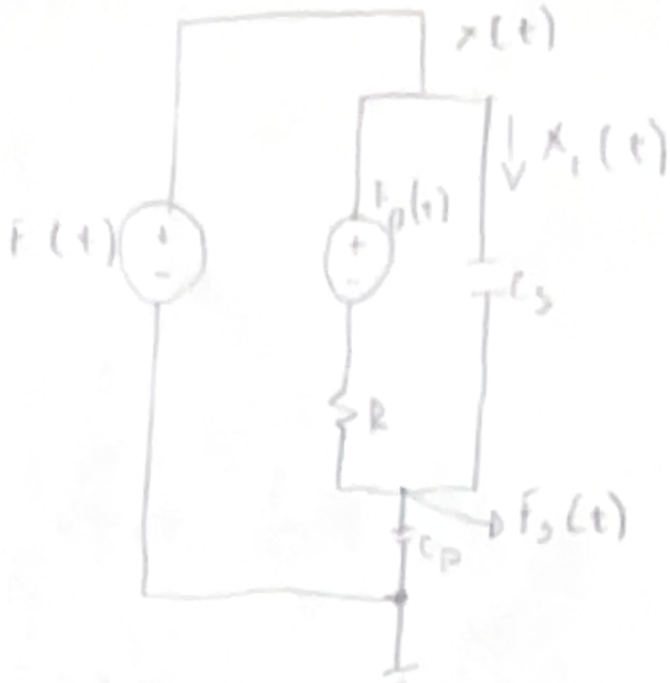
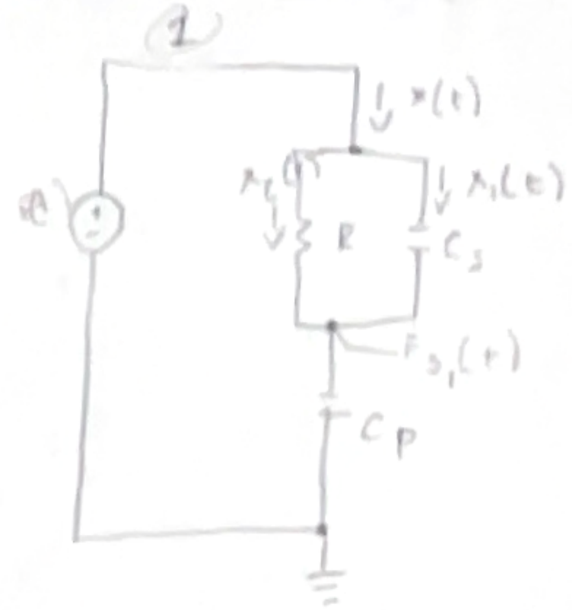
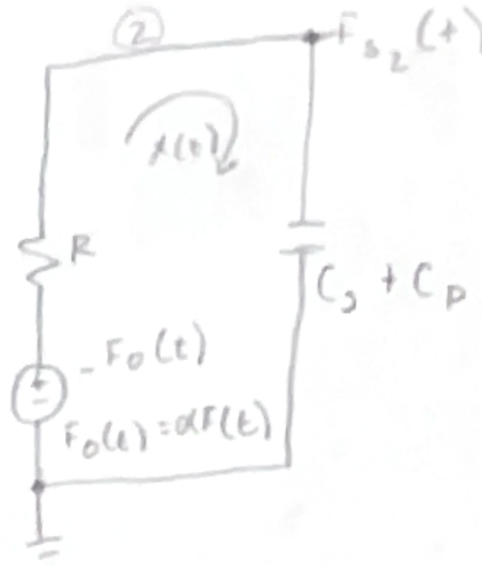
Práctica 3: Modelado y control del sistema musculoesquelético









Análisis del circuito por superposición

Circuito 1: Apagando $F_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s_1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p \frac{dF_{s_1}(t)}{dt} = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt} + \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p s F_{s_1}(s) = C_s s [F(s) - F_{s_1}(s)] + \frac{F(s) - F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) = C_s s F(s) - C_s s F_{s_1}(s) + \frac{F(s)}{R} - \frac{F_{s_1}(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2: Apagando F(t)

$$-\alpha F(t) = R_x(t) + \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$Fs_2(t) = \frac{1}{(C_p + C_s}\int x(t)\,dt$$

$$-\alpha F(s) = RX(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$Fs_2(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{x(s)}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{Fs_2(s)}{F(s)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{x(s)}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$Fs_2(s) = \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) + F_{s_1}(s) + F_{s_2}(s) = \frac{C_s R_s + 1}{R(C_p + C_s)s + 1}F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R_s + 1 - \alpha}{R(C_p + C_s)s + 1}F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R_s + 1 - \alpha}{R(C_p + C_s)s + 1}$$